C6: Video Analysis [MO83391]

Multi-Target Multi-Camera Tracking

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Agenda

Júlia

01 Dataset Overview: CityFlow

02 Method Pipeline

03 Algorithm: Cross-Camera Candidate Search

04 Car Matching

05 Quantitative Results

06 Qualitative Results

07 Conclusions

Dataset Overview: CityFlow

Júlia

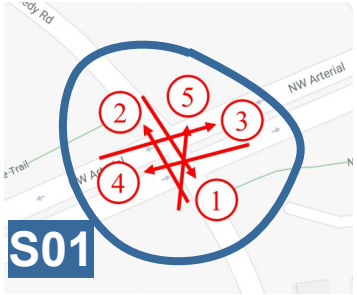
Dataset Overview

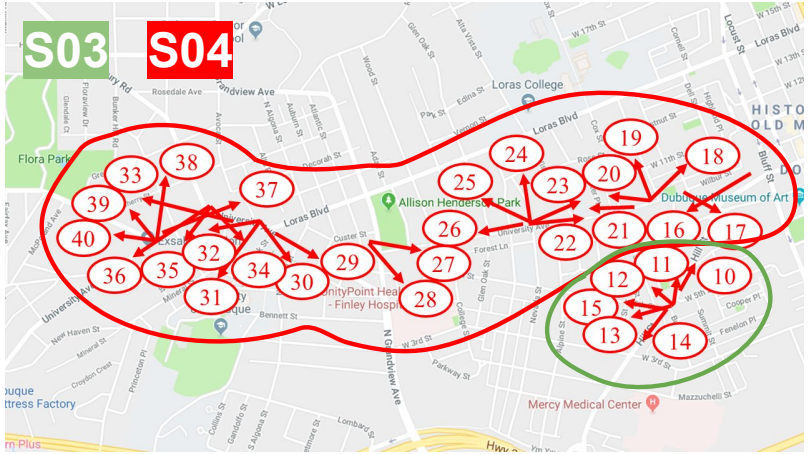
- City-scale multi-camera vehicle tracking
- **6 sequences - 40 cameras - 10 intersections**
- ~3 hours of synchronized videos
- ~230K annotated bounding boxes

Our Setup

We focus on sequences: **S01, S03, S04**

- Train: S01, S04
- Test: S03





Available Data per Camera

- Homography: from real-world (GPS) coordinates to image space
- Temporal synchronization between cameras
- Cross-camera vehicle identities (MTMC annotations)

Zheng Tang, et al., CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification CVPR 2019

Method Pipeline

Júlia

Single-Camera Stage

Multi-Camera Stage

Maximum Overlap from Week 2

Car Tracking

Compare candidate pairs and decides if they correspond to the same vehicle.

Car Matching NEW

Car Detection

Faster-RCNN from Week 2

Cross-Camera Candidate Search NEW

Select candidate tracks in other cameras that may belong to the same vehicle.

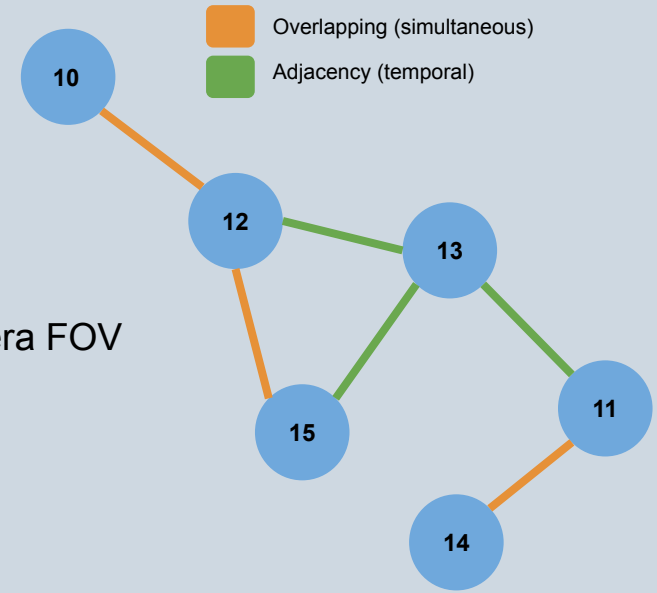
Algorithm: Cross-Camera Candidate Search

Adrián

Algorithm Overview

1. **Connect** all cameras that are adjacent or overlapping
2. **Iterate** through every frame
   - 2.1. **Iterate** through every camera
     - 2.1.1. **Register** new cars
     - 2.1.2. **Mark** cars as active
   - 2.2. **Iterate** through active cars
     - 2.2.1. **Check** if they appear on another camera
     - 2.2.2. **Otherwise, check** if they are about to exit the camera FOV
   - 2.3. **Iterate** through overlapping or exiting cars
     - 2.3.1. **Mark** close cars in case of overlapping scenario
     - 2.3.2. **Sort** by distance
     - 2.3.3. **Match** cars using matcher

Precomputed:
- **MultiObjectSingleCamera predictions**

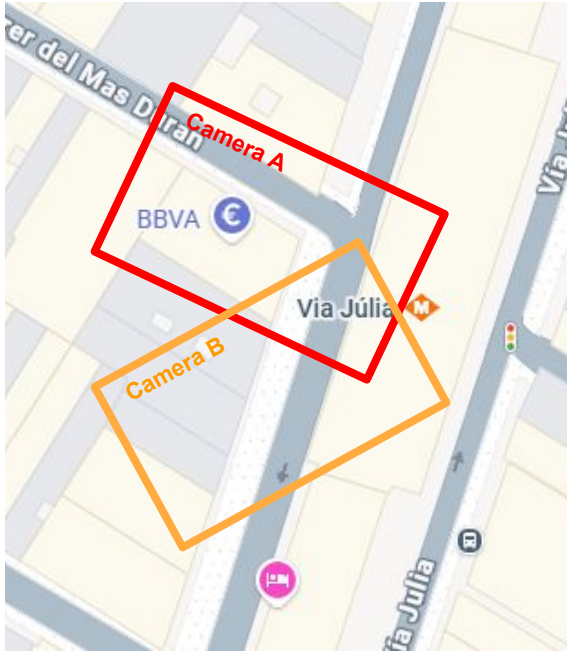


Algorithm: Cross-Camera Candidate Search

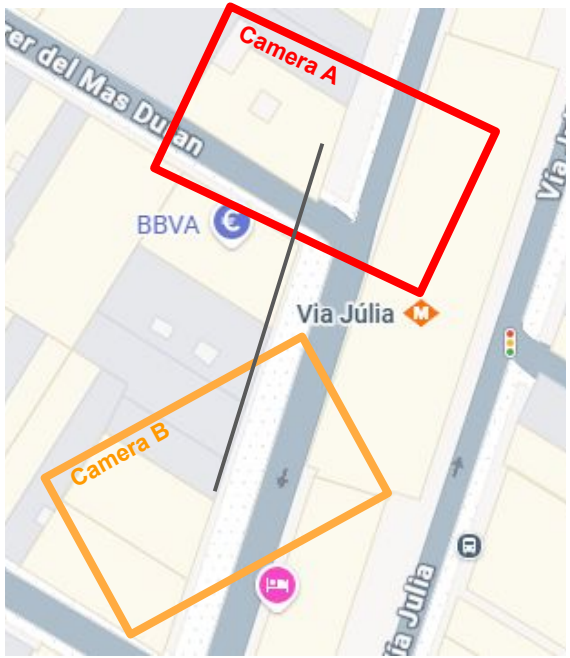
Adrián

Camera Connections

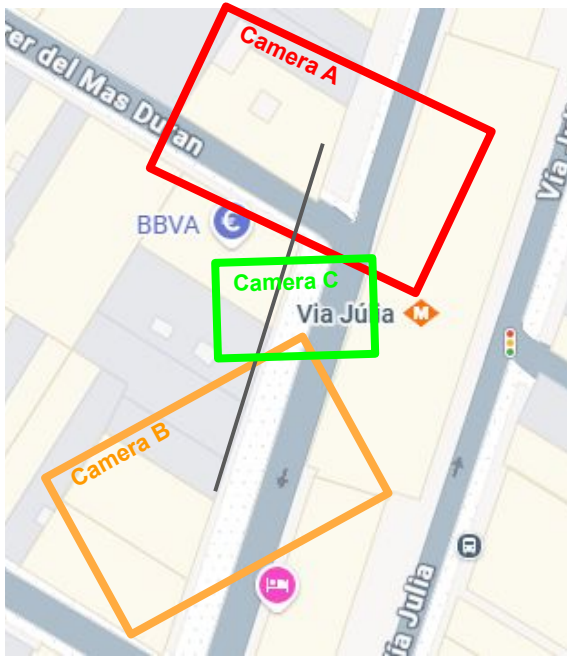
Overlapping Cameras



Adjacent Cameras



Non-Adjacent Cameras



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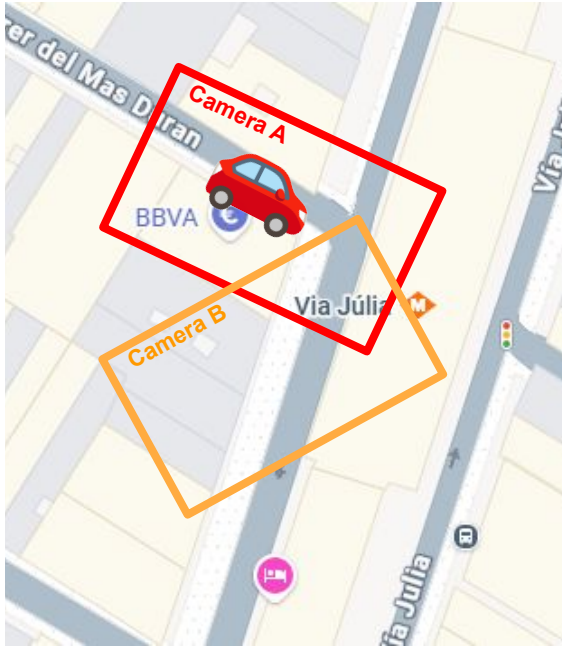
Algorithm: Cross-Camera Candidate Search

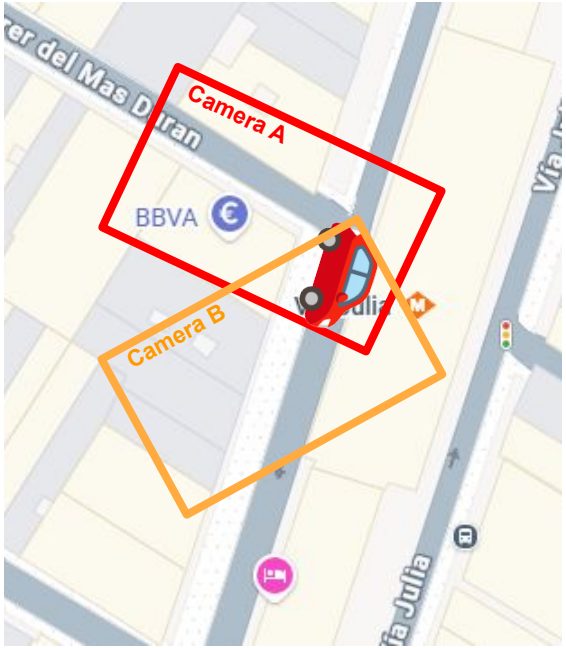
Adrián

Car State

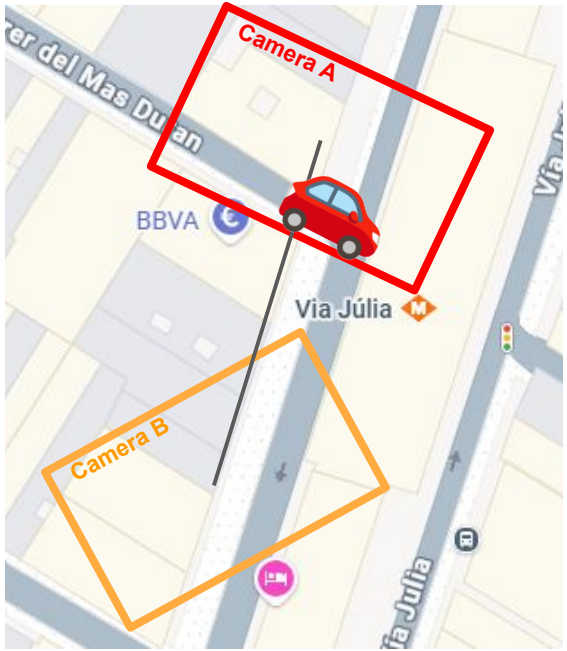
Nothing Car



Overlapping Car



Exiting Car



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Algorithm: Cross-Camera Candidate Search

Adrián

Car Matching Conditions

Car A: Source car (is looking for a match)
Car B: Target car (is a candidate to Car A)

If **Car A** already has a match in the Target camera

We **stop looking** into that camera.

If **Car B** already has a match in the Source camera

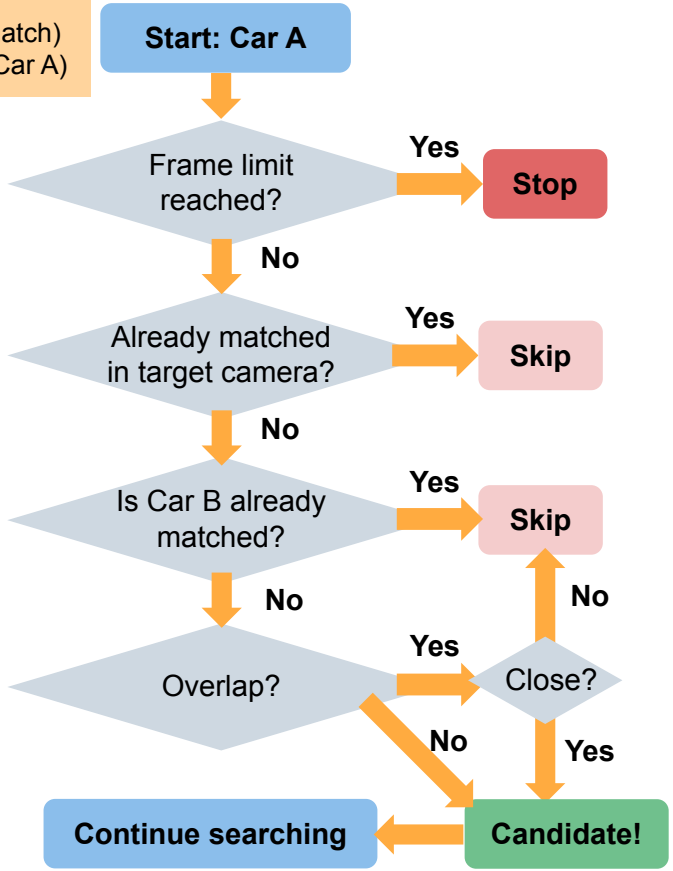
We **skip** Car B.

If **Car A** is considering an overlap

They should be **close** in real world.

If **Car A** has been looking for N frames

We **stop looking** for matching cars.



Car Matching

Adrián

Vehicle Re-Identification Model

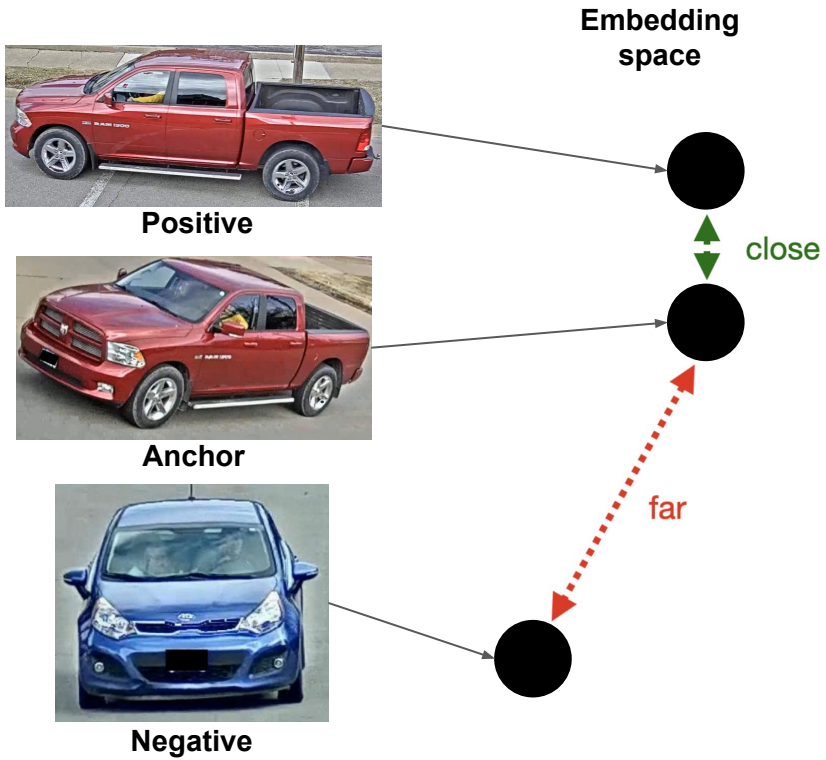
Vehicle **Re-Identification (ReID)** aims to determine whether two images of vehicles correspond to the **same physical car**, even if they appear in **different cameras, viewpoints, or lighting conditions**.

The model learns a **visual embedding space** where:

- Images of the **same vehicle are close**
- Images of **different vehicles are far apart**

These embeddings can then be used to **associate vehicle tracks across cameras** though **cosine similarity**.

Are these two images of the same car?



Car Matching

Marina

Model architecture & Training Strategy

CNN Backbone
DenseNet-121
Pretrained on ImageNet

↓

Embedding Layer
Trained from scratch

↓

Classification Head
Trained from scratch
Only used for training

1. Cross-entropy Loss (Identity Classification)

Encourages the model to correctly classify each vehicle identity. This increases **inter-class separability**.

2. Batch-Hard Triplet Loss

Triplet loss enforces that embeddings satisfy:

d(anchor, positive) < d(anchor, negative)

Where:

- **anchor**: image of a vehicle
- **positive**: another image of the same vehicle
- **negative**: image of a different vehicle

Final Training Objective

Total loss: $L = L_{cross_entropy} + L_{triplet}$

This combination is commonly used in **vehicle and person ReID models** and was also evaluated in the CityFlow benchmark.

W. Tang, Z. Liu, Y. Yu, B. Shi, Y. Xu, J. Li, Y. Wang. CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification. CVPR 2019

Car Matching

Marina

Track Representation

Vehicles are first **detected and tracked** within each camera.

For each detection:

Vehicle Crop → **ReID Model** → **Embedding**

A single descriptor per track is obtained by **averaging embeddings across frames**:

$$e_{track} = \frac{1}{N}\sum_{i=1}^{N} e_i$$

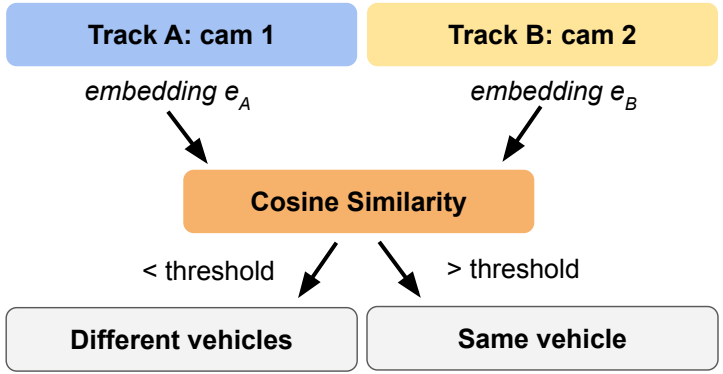
Welford algorithm used for efficiency

This aggregation produces a **robust representation of the vehicle appearance**, reducing noise caused by occlusions, motion blur, or detection errors.

To associate vehicles across cameras, track embeddings are compared using **cosine similarity**.

$$d = 1 - (e_A \cdot e_B)$$

Tracks with **high similarity** are considered to correspond to the **same vehicle identity**.



Car Matching

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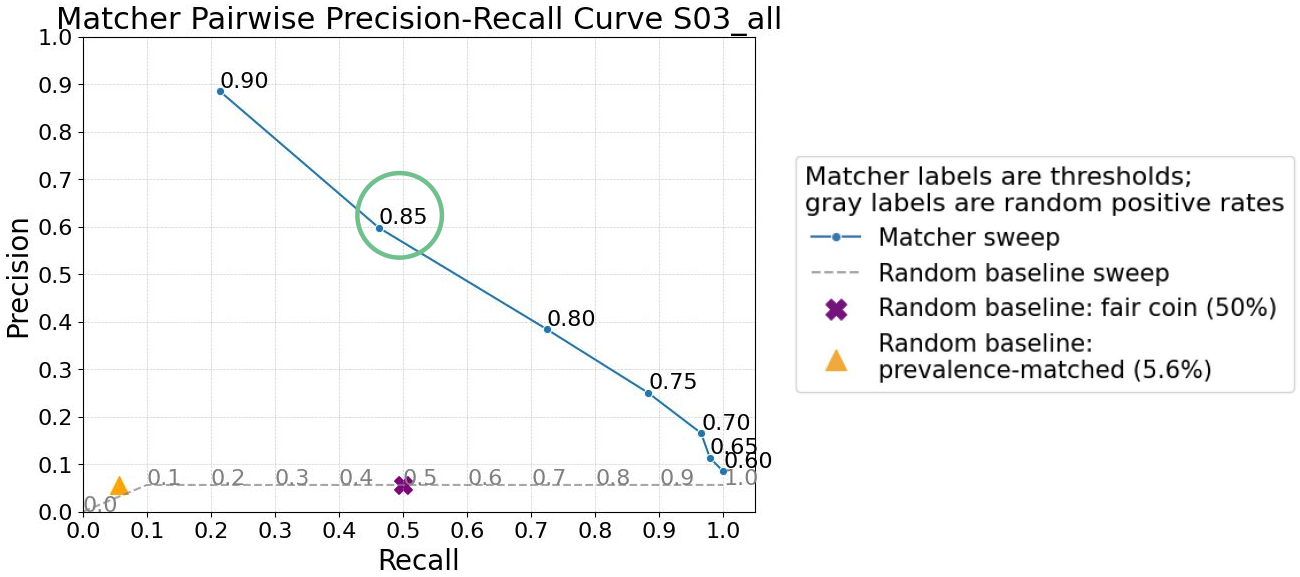
ReID - Threshold sweep

Matcher evaluated independently using GT tracks



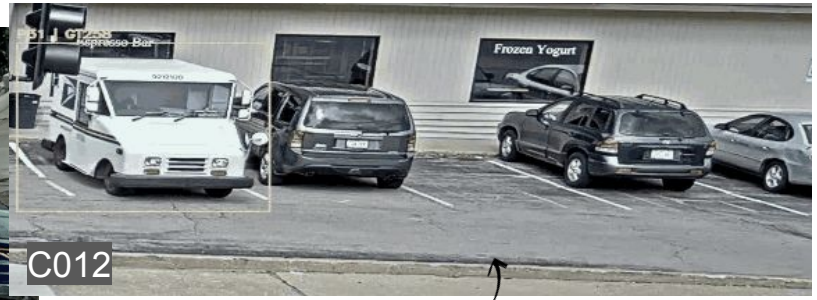
Clear precision-recall trade-off across thresholds

Threshold 0.85 provides best balance for matching

Car Matching

Marina

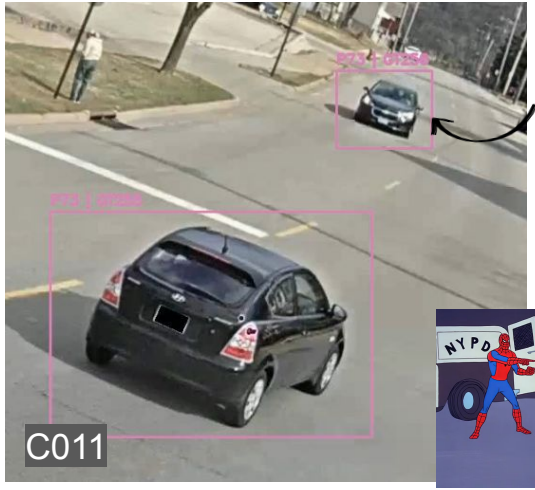
ReID - Qualitative results



Different IDs assigned

Same ID assigned

These are the same car



In some cases we had 2 vehicles in the same frame classified as the same vehicle

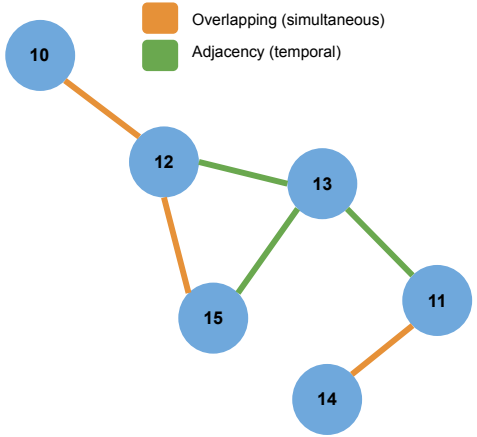
Quantitative Results

Lore

S03 Test Sequence

MTMC

Sequence	IDF1
S03	**37.11**



MTSC

Camera	IDF1
c010	30.88
c011	**1.43**
c012	**96.35**
c013	44.04
c014	20.80
c015	**0.00**

Reminder

IDF1 measures how consistently we maintain the identity of each vehicle over time.

Note

It took around 19 min to evaluate the whole S03 sequence.

Qualitative Results

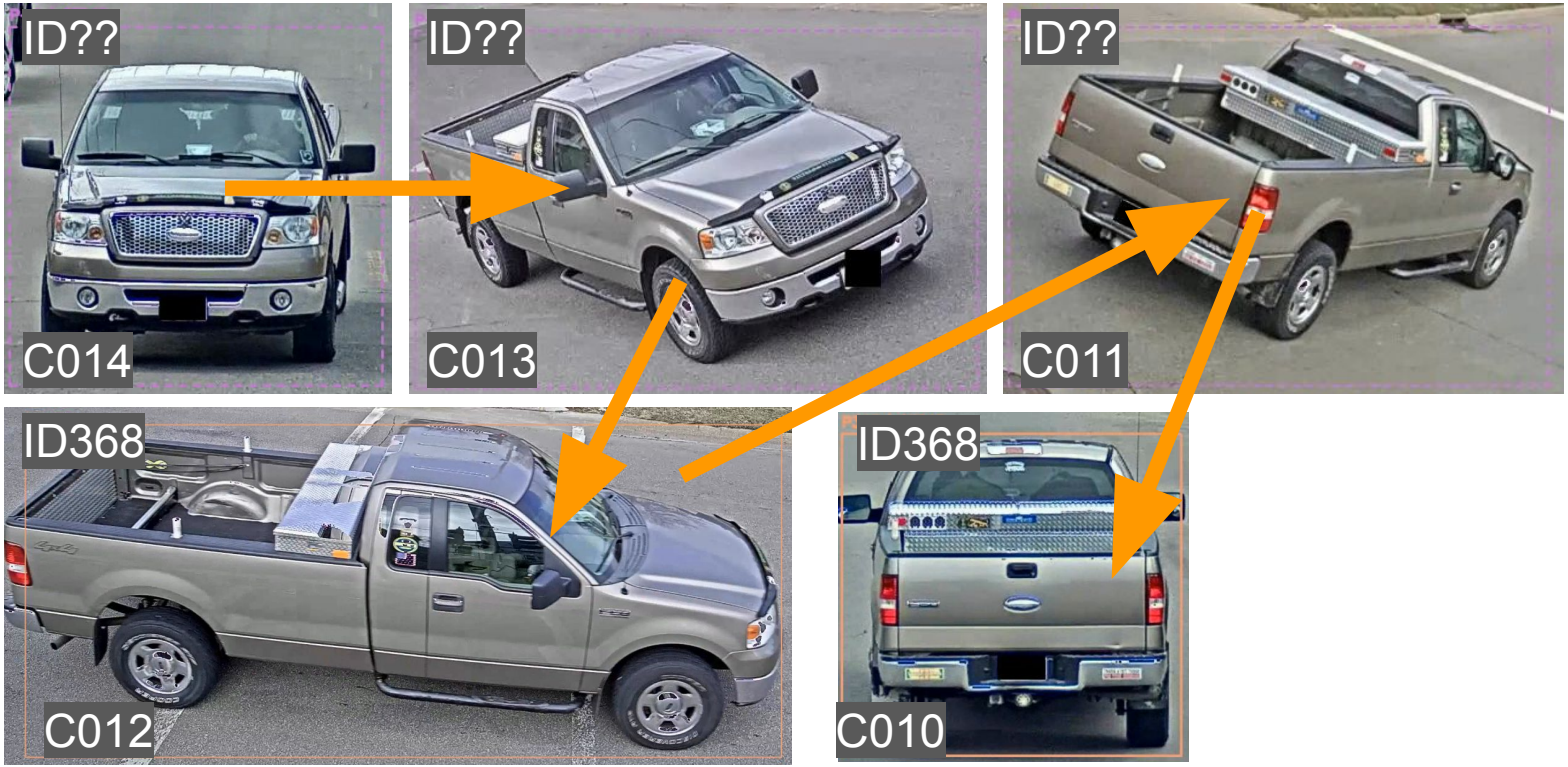
Lore

Incorrect Cross-Camera Match

Qualitative Results

Lore

Gap in the Tracking Across Cameras



Qualitative Results

Lore

Effect of Spatial Filtering (source car from C012)

Conclusions

Lore

Algorithm performance

- The proposed pipeline shows potential for multi-camera tracking.
- However, in its current form, the performance is limited, especially in cross-camera association.
- Errors in detection and tracking can propagate and negatively impact cross-camera matching.

Observed Problems

- Different frame rates make it harder to synchronize multiple cameras in real time.
- Simplifying connections can lead to unreal restrictions when filtering.
- Appearance-based matching alone is not be sufficient in challenging scenarios.

Key Insights

- Strong single-camera tracking does not guarantee good multi-camera performance.
- Combining spatial, temporal, and appearance cues is essential for robust MTMC tracking.

Possible Improvements

- Adaptive distance threshold.
- Use speed as heuristic (temporal and spatial).
- Use road connections to connect cameras.
- Wait for M frames before trying to match cars.
- More weight to new embeddings.
- Car reappearances.

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